

Analysis of Scientific Data



Lecture 12

Nonparametric Methods

Darwin (1902)

An experiment by Darwin involved a comparison of the growth of cross-fertilised and self-fertilised plants, with pairs of plants grown together in pots to eliminate other factors.

THE EFFECTS OF CROSS AND SELF FERTILISATION

IN THE
VEGETABLE KINGDOM.

By CHARLES DARWIN, M.A., F.R.S.,
ETC.

NEW YORK:
D. APPLETON AND COMPANY,
73 FIFTH AVENUE.
1902.

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ZEA MAYS.

CHAP. VI.

also anemophilous, or is fertilised by the wind; and of such plants only the common beet had been tried. Some plants were raised in the greenhouse, and were crossed with pollen taken from a distinct plant; and a single plant, growing quite separately in a different part of the house, was allowed to fertilise itself spontaneously. The seeds thus obtained were placed on damp sand, and as they germinated in pairs of equal age were planted on the opposite sides of four very large pots; nevertheless they were considerably crowded. The pots were kept in the hothouse. The plants were first measured to the tips of their leaves when only between 1 and 2 feet in height, as shown in the following table:—

TABLE XXVII.
Zea mays.

No. of Pot.	Crossed Plants.	Self-fertilised Plants.
I.	Inches. 23½ 12 21	Inches. 17½ 20½ 20
II.	22 19½ 21½	20 18½ 18½
III.	22½ 20½ 18½ 21 23½	18½ 15½ 18½ 18 16½
IV.	21 22½ 23 12	18 12½ 15½ 18
Total in inches.	302·88	263·63

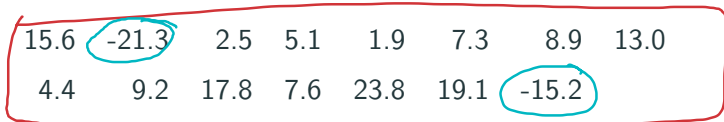
The fifteen crossed plants here average 20·19, and the fifteen self-fertilised plants 17·57 inches in height; or as 100 to 87. Mr. Galton made a graphical representation, in accordance with the method described in the introductory chapter, of the above

male flowers usually shed their pollen before the female flowers are mature: 'Monatsbericht der

K. Akad. Berlin, Oct. 1872, p. 743.

Darwin's Data

In centimetres, the differences in heights between the cross- and self-fertilised plants in each of 15 pots are



15.6	-21.3	2.5	5.1	1.9	7.3	8.9	13.0
4.4	9.2	17.8	7.6	23.8	19.1	-15.2	

Darwin's hypothesis was that cross-fertilised plants would grow more vigorously than those that had been self-fertilised.

t-test

Does a t -test give any evidence that cross-fertilised plants have grown more than self-fertilised plants?

Let μ = mean difference in growth between cross- and self-fertilised plants

Test $H_0: \mu = 0$ vs $H_1: \mu > 0$

$\bar{x}$ increase

s much bigger

Observe $\bar{x} = 6.6467$, $s = 11.99$, $n = 15$

$$p\text{-value} = P(\bar{X} \geq 6.6467)$$

$$= P\left(Z \geq \frac{6.6467 - 0}{11.99/\sqrt{15}}\right)$$

$$= P(Z \geq 2.147)$$

$$= 0.0249$$

decrease



parametric test

assumes data follow a specified distribution

non parametric test:

makes no strict assumptions about the population distribution

moderate evidence that the mean growth is higher with cross-fertilised plants

Poll Question

Suppose the first observation was incorrectly entered as '156' instead of '15.6'.

How will this affect the t -test result?

1. Stronger evidence that cross-fertilised plants grow more
2. Weaker evidence that cross-fertilised plants grow more
3. Not sure

higher p -value

Sign Test

Count the number of positive differences.

What is the distribution of this count, X , if there was no underlying difference? $X \sim \text{Binomial}(15, 0.5)$

Observe $x = 13$ from $n = 15$.

$$p\text{-value} = P(X \geq 13)$$

$$= P(X=13) + P(X=14) + P(X=15)$$

$$= 0.00369$$

$\text{dbinom}(13, 15, 0.5)$

$\text{binom.test}(13, 15, \text{alternative} = "greater")$

Strong evidence that cross-fertilised plants are taller than self-fertilised plants.

What **parameter** is the sign test implicitly testing for? **median**

Let m = population median

Test $H_0: m = 0$ vs $H_1: m > 0$

Signed-Rank Test

The sign test is robust but very wasteful, leading to low power.

The **signed-rank test** improves on this by taking into account the relative magnitude of the values.

11 15.6	14 -21.3	2 2.5	4 5.1	1 1.9	5 7.3	7 8.9	9 13.0
4.4	9.2	17.8	7.6	23.8	19.1	-15.2	
3 (11)	8 (14)	12 (2)	6 (4)	15 (1)	13 (5)	10 (7)	

① Rank all observations from smallest to largest, ignoring the sign.

② Calculate S = sum of ranks from positive differences

$$S = 11 + 2 + 4 + 1 + 5 + 7 + 9 + 3 + 8 + 12 + 6 + 15 + 13$$
$$= 96$$

③ $p\text{-value} = P(S \geq 96) = P(Z \geq \frac{96 - 60}{\sqrt{310}}) = P(Z \geq 2.0447)$

$sd(S)$

$$= 0.02044$$

$$n=15, S=0,1,2,\dots,120$$

moderate evidence

What is the null distribution of S ?

Write $S = S_1 + S_2 + \dots + S_n$ where S_i has distribution $Pr(S_i) = \begin{array}{c|cc} S_i & 0 & i \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}$

By the Central Limit Theorem, when n is large,

$$S \overset{\text{approx}}{\sim} N(E(S), \text{sd}(S))$$

$$S \sim N(60, \sqrt{310})$$

$$E(S_i) = \frac{1}{2} \times 0 + \frac{1}{2} \times i = \frac{1}{2}i$$

$$\text{Var}(S_i) = \frac{1}{2}(0 - \frac{1}{2}i)^2 + \frac{1}{2}(i - \frac{1}{2}i)^2 = \frac{1}{2}(\frac{1}{4}i^2) + \frac{1}{2}(\frac{1}{4}i^2) = \frac{1}{4}i^2$$

$$\text{So } E(S) = E(S_1) + E(S_2) + \dots + E(S_n)$$

$$= \frac{1}{2} \times 1 + \frac{1}{2} \times 2 + \dots + \frac{1}{2} \times n$$

$$= \frac{1}{2} (1 + 2 + \dots + n)$$

$$= \frac{1}{2} \frac{n(n+1)}{2}$$

$$= \frac{n(n+1)}{4} \quad \frac{15(16)}{4} = 60$$

$$\text{Var}(S) = \text{Var}(S_1) + \text{Var}(S_2) + \dots + \text{Var}(S_n)$$

$$= \frac{1}{4} \times 1^2 + \frac{1}{4} \times 2^2 + \dots + \frac{1}{4} \times n^2$$

$$= \frac{1}{4} (1^2 + 2^2 + \dots + n^2)$$

$$= \frac{n(n+1)(2n+1)}{24} \quad \frac{15 \times 16 \times 31}{24} = 310$$

$$\text{sd}(S) = \sqrt{\frac{n(n+1)(2n+1)}{24}}$$

Poll Question

If we want to test whether cross-fertilised and self-fertilised plants grow at different rates, what would the p -value be?

(Recall the p -value from our previous test was 0.0206.)

~~2x 0.0206~~

1. 0.0206

2. 0.0412

3. 0.0103

4. 0.9588

(Wilcoxon) Rank-Sum Test

Wilcoxon (1945) describes a nonparametric test for comparing two independent samples.

Is there a difference in ratings between the two brands?

Brand X ratings: 3, 4, 2, 6, 2, 5

Brand Y ratings: 9, 7, 5, 10, 6, 8

H_0 : the distribution of ratings for brand A and brand B is the same.

H_1 : " " " " " " " " " not the same.

① Rank all observations, ignoring signs and groups.

If numbers tie, take the average rank.

② Calculate W = sum of ranks from group 1 (Brand X).

$$W = 3 + 4 + 1.5 + 7.5 + 1.5 + 5.5 = 23$$

$$\text{Under } H_0, W \overset{\text{approx}}{\sim} N\left(\overset{E(W)}{\frac{n_1(n_1+n_2+1)}{2}}, \overset{sd(W)}{\sqrt{\frac{n_1 n_2 (n_1+n_2+1)}{12}}}\right)$$

$$n_1 = n_2 = 6$$
$$E(W) = \frac{6 \times 13}{2} = 39, \quad sd(W) = \sqrt{\frac{6 \times 6 \times 13}{12}} = \sqrt{39}$$

③ (one-sided)
p-value = $P(W \geq 23)$

$$= P\left(Z \geq \left| \frac{23 - 39}{\sqrt{39}} \right| \right)$$

$$= P(Z \geq | -2.56205 |)$$

$$= 0.0052$$

Two-sided p-value = 0.0104

Moderate evidence that the distribution of the ratings of the two brands are different.

Summary of Nonparametric Tests

Nonparametric tests are useful when standard assumptions are not satisfied, or when your data is already ranked (such as ordinal data).

Various nonparametric tests exist, such as

- Sign test (for one-sample ^{t-test} or paired data)
- Signed-Rank Test (for one-sample ^{t-test} or paired data)
- Wilcoxon Rank-Sum Test or Mann-Whitney Test (comparing two groups) ^{2-sample t-test (pooled)}
- Kruskal-Wallis Test (comparing ^{1-way ANOVA} more than two groups)
- Spearman's Correlation (testing for association)